

Fundamental Deformation and Recovery Behaviors of Ni-Ti-Nb Shape Memory Alloy*

Masaki KUSAGAWA**, Toshiya NAKAMURA***
and Yasuhide ASADA****

Fundamental deformation and deformation-recovery behaviors of Ni-Ti-Nb shape memory alloy were experimentally investigated, especially their dependency on test temperature. Monotonic tension tests and deformation-recovery tests were conducted at low and high temperatures ranging from 253 K to 473 K. Stress-strain response above 323 K is quite different from that below the temperature. The yield stress shows the positive temperature-dependency below 323 K and the opposite trend is observed above this temperature. From this result, it is estimated that the dominant mechanism of inelastic deformation is the plastic deformation above 323 K and the pseudo-elastic one below it. The temperature at which strain recovery starts strongly depends on the maximum prestrain and prestraining temperature. Residual strain is always observed in the deformation-recovery tests. Also it is found that the larger recovery strain can be obtained for the larger prestrain and at the lower prestraining temperature.

Key Words: Material Testing, Inelasticity, Tensile Properties, Ni-Ti-Nb Shape Memory Alloy, Stress Induced Martensite Transformation, Shape Recovery, Residual Strain

1. Introduction

It is more than forty years since the shape memory effect was found in some alloys. Many material properties and mechanisms have been investigated since then. A shape memory alloy (SMA) shows interesting deformation behaviors: pseudo-elasticity and shape memory effect. There are already many engineering applications using these material properties. Good reviews are given by Tanaka et al.⁽¹⁾ and Fremont et al.⁽²⁾ Another type of application can be found recently that Ni-Ti-Nb SMA is used as a structural material in repairing the incore monitoring

system of a nuclear power plant⁽³⁾.

As a structural material the phase of SMA should be the austenite during service duration because the yield stress of the martensite is quite small⁽⁴⁾. The service temperature should be higher than M_s at which the martensite transformation starts but the pre-deformed SMA device must be stored at the temperature below A_s , at which the shape recovery starts. Thus, in such an application, wide temperature hysteresis of phase transformation is desirable. Ni-Ti-Nb SMA was developed based on the well-known Ni-Ti SMA to satisfy the engineering requirement mentioned above^{(4),(7)}. Indeed, $M_s = 219.7$ K (-53.3°C) for the Ni-Ti-Nb SMA used in the present study and the material is designed to be austenite within the temperature range from room temperature to 288°C which is the service temperature of LWR plant.

The present material consists of two different phases: Ni-Ti matrix and Nb rich particle⁽⁴⁾⁻⁽⁷⁾. Piao et al.⁽⁵⁾ and Zhang et al.⁽⁶⁾ reported the following deformation mechanisms of the material. The phase at low temperatures is martensite and the yield stress is higher for the lower temperature. The yielding at such a low temperature is due to the motion of

* Received 2nd February, 2000. Japanese original: Trans. Jpn. Soc. Mech. Eng., Vol. 65, No. 636, A (1999), pp. 1734-1740 (Received 14th October, 1998)

** Osaka Gas, 4-1-2 Hirano, Chuo-ku, Osaka 541-0046, Japan

*** Department of Mechanical Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan. E-mail: nakamt@mech.t.u-tokyo.ac.jp

**** Central Research Institute of Electric Power Industry, 2-11-1 Iwato-kita, Komae, Tokyo 201-8511, Japan

martensite boundaries or twinning. This deformation brings no damage and then the deformation disappears by heating (shape memory effect). At the medium temperature the material shows a sharp yielding behavior but the strain hardening is quite small. This "plastic" deformation comes from the stress induced martensite transformation (SIM) thus the "yield" stress at which SIM starts is higher for higher temperature. At the higher temperature the plastic deformation due to slip starts in the soft Nb rich phase before SIM starts and it reduces the shape memory effect. When the temperature is very high, the plastic deformation of the austenite starts which gives the limitation of shape memory effect. These features are quite similar to those of well-known Ni-Ti SMA⁽⁵⁾ but some differences are found that come from the plastic deformation of Nb rich phase. Furthermore, Duerig⁽⁴⁾ reported that the residual strain is usually observed in the shape recovery behavior of Ni-Ti-Nb SMA and that the A'_s temperature at which the shape recovery starts becomes higher for the larger prestrain.

As mentioned above Ni-Ti-Nb SMA shows interesting properties different from well-known Ni-Ti SMA. But when considering the application as a structural material, further experimental investigation should be made on the deformation properties coming from SIM, shape memory effect. In the present study monotonic tension tests and shape recovery tests were conducted to investigate the fundamental deformation/recovery behavior of the material including their dependencies on temperature.

2. Experimental Procedure

2.1 Material and specimen

The material tested in the present study is a Ni-Ti-Nb SMA whose chemical composition is shown in Table 1. The phase transformation temperatures were measured by the differential scanning calorimeter (DSC) as shown in Table 2. From Table 2 one can find that the phase transformation temperatures are low and that the material is austenite at the room temperature. It is noted that the measured results in Table 2 almost coincide with those reported by literatures⁽⁴⁾⁻⁽⁷⁾.

Test specimens were machined out from a SMA ring (outer diameter=75 mm, inner diameter=53.5 mm, length=80 mm). The specimen is a solid cylinder whose diameter is 5 mm, length is 78 mm with straight section of 15 mm. The gage length (GL) is 12 mm.

2.2 Experimental equipment

Electro-hydraulic closed loop system is used in the present study. The configuration of the entire system is shown in Fig. 1. As seen in the figure, the

Table 1 Chemical composition, wt%

Ni	Ti	Nb
47.6	36.2	15.8

Table 2 Phase transformation temperature K(°C)

M_f	M_s	A_s	A_f
184.6	219.7	247.5	277.3
(-88.4)	(-53.3)	(-25.5)	(4.3)

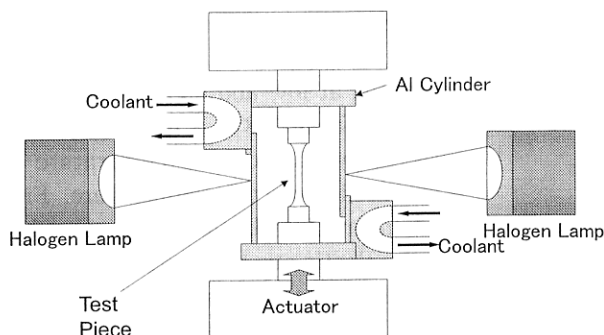


Fig. 1 Test equipment

specimen is covered by a black cylinder made of Al alloy to attain a uniform temperature distribution. Two sets of Halogen lamps are used for heating. The maximum temperature difference measured in GL was 2 K at 473 K. A cooling system was installed for the experiments at low temperatures. The maximum temperature difference in GL was 1 K at 253 K.

The elongation was measured by a hand made displacement sensor of DTF type. It was confirmed that the drift due to temperature change is small enough to be neglected during the experiments.

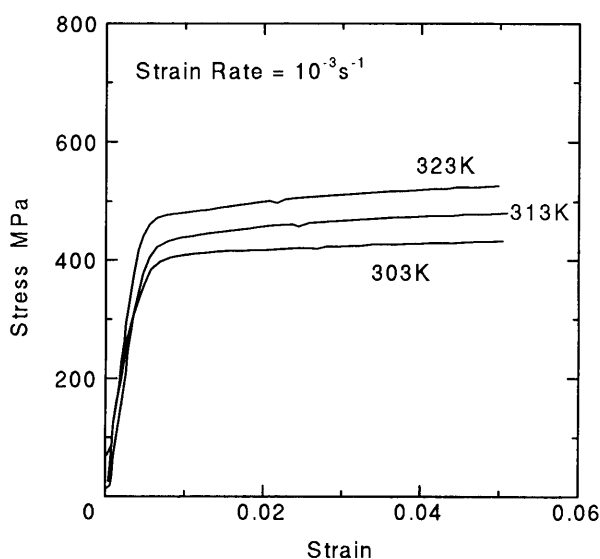
3. Monotonic Tension Tests

Monotonic tension tests were conducted at 303, 313, 323, 373, 423, and 473 K (30, 40, 50, 100, 150, 200°C), with a strain rate of 10^{-3} s^{-1} . The maximum strain was 0.05. The obtained stress-strain curves are shown in Figs. 2(a), (b). It can be observed that the stress response above 323 K (50°C) is quite different from that below 323 K. Figure 2(a) shows that the yield stress is larger for higher temperatures below 323 K but the opposite tendency is found at the higher temperature as seen in Fig. 2(b). The strain hardening is not so significant below 323 K but is so at the higher temperatures.

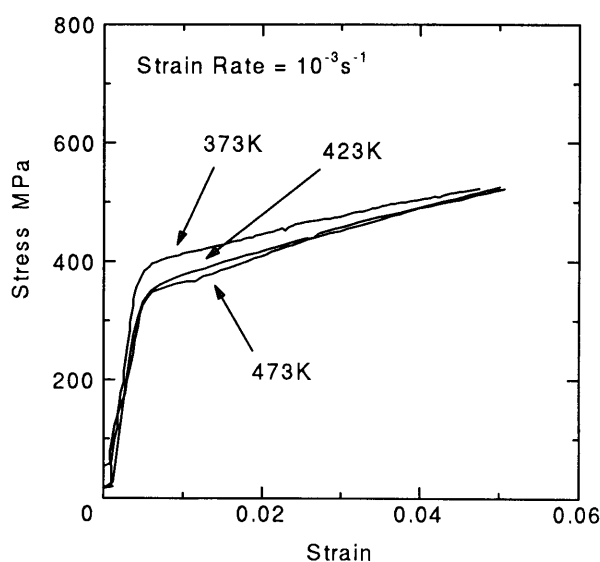
4. Strain Recovery Tests

4.1 Determination of strain rate

Monotonic tension tests were conducted with



(a) Stress-strain curves at 303, 313, 323 K



(b) Stress-strain curves at 373, 423, 473 K

Fig. 2 Stress-strain curves from monotonic tension tests

different strain rates of 10^{-3} s^{-1} and 10^{-4} s^{-1} , with the maximum strain of 0.07 at room temperature. The responses of stress and specimen temperature are shown in Fig. 3. As is easily seen in the figure, the specimen temperature rises during tensile deformation. It is significant (7.7 K) for the faster strain rate of 10^{-3} s^{-1} but very little (1.8 K) for the slower strain rate of 10^{-4} s^{-1} . This temperature rise comes from the latent heat of the phase transformation. The rate dependent deformation of Cu-Al-Ni SMA was studied by Otsuka et al.^{(8),(9)} and rate dependent constitutive modeling was discussed by Abeyaratne et al.^{(10)–(14)}

Since the effect of this temperature rising upon strain recovery is not clear at this time, the strain rate of 10^{-4} s^{-1} was selected for the following experiments

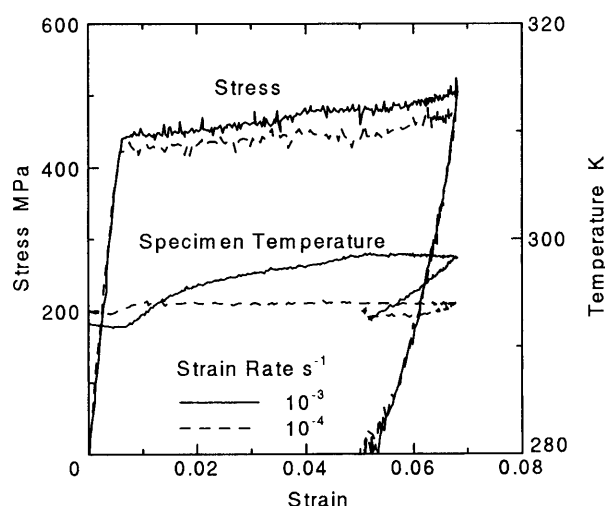


Fig. 3 Rate-dependence of stress and temperature

to investigate the strain recovery behavior.

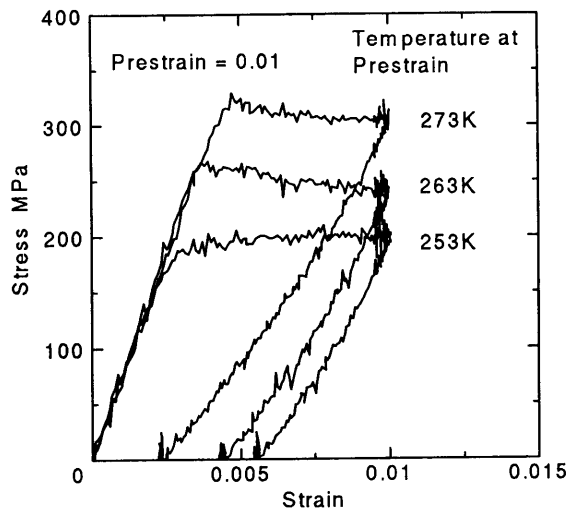
4.2 Test condition

The effects of the prestrain temperature and the magnitude of prestrain on the strain recovery were investigated. The prestrain temperatures are 253, 263, 273 K (-20 , -10 , 0°C) and prestrain (nominal strain) were 0.01, 0.03, 0.07. The test procedure is as follows: (1) cool the specimen down, and after several hours, (2) set the zero point of the displacement sensor, (3) stretch the specimen with strain rate of 10^{-4} s^{-1} up to the prescribed strain, (4) unload until zero stress, (5) heat up the specimen with a heating rate of 20 K/min up to 473 K (200°C), (6) keep the temperature for 30 min to observe the strain recovery, and then (7) the specimen was cooled down to the pre-straining temperature.

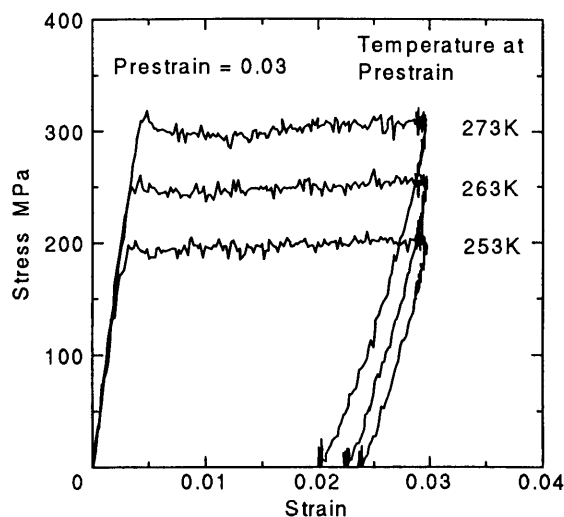
4.3 Test result

4.3.1 Stress strain behavior Stress strain curves are shown in Figs. 4(a), (b) and (c). It should be noted that the strain in the figures are the true strain $\epsilon = \ln(1 + \epsilon_n)$, where ϵ_n is the controlled nominal strain. Some interesting behaviors are pointed out at these low temperatures that are not observed above room temperatures. A “sharp” yielding and the subsequent reduction of stress is observed. Strain hardening takes place after this reduction of stress but it is not significant. The yield stress and the flow stress are smaller for the lower temperature. Figure 4(c) shows a large hardening after 0.05 strain. Finally, the slope of the unloading portion is smaller than that (Young’s modulus) of the first loading portion.

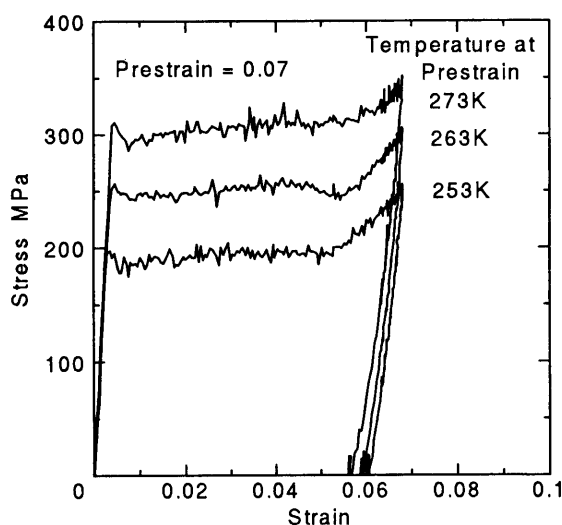
4.3.2 Strain recovery Heating after unloading brought about the recovery of strain. The rising temperature reduced the strain (strain recovery) but simultaneous thermal expansion took place. Thus the



(a) 1% prestrain



(b) 3% prestrain



(c) 7% prestrain

Fig. 4 Stress-strain curves from deformation recovery tests

residual strain was measured after the specimen was cooled down to the prestraining temperature. In all the cases the perfect recovery was not established and residual strain was always observed. Details will be discussed in the next section.

5. Discussion

5.1 Stress strain response and effects of temperature

The relation between the measured Young's modulus and specimen temperature is shown in Fig. 5. The Young's modulus is larger for higher temperature. This trend is opposite to the usual case of steels or alloys, but coincides with the case of Ni-Ti SMA⁽¹⁾.

Figure 5 also shows the relation between the yield stress and specimen temperature. It can be pointed out that there is a positive linear relation below 323 K (50°C). This is a well known feature of the stress induced martensite transformation (SIM) and thus the "inelastic" deformation below 323 K at least is the pseudo-elastic due to SIM. At the higher temperature, 323, 373, 423 and 473 K (50, 100, 150, 200°C) the yield stress shows negative relation with the temperature. This trend coincides with the usual case of steels or alloys, which indicates that the "inelastic" deformation at these high temperatures is the plastic deformation of the austenite phase.

5.2 Strain recovery

5.2.1 Temperature at which strain recovery starts As mentioned in the subsection 4.3.2, residual strain was observed in all the tests. Figure 6

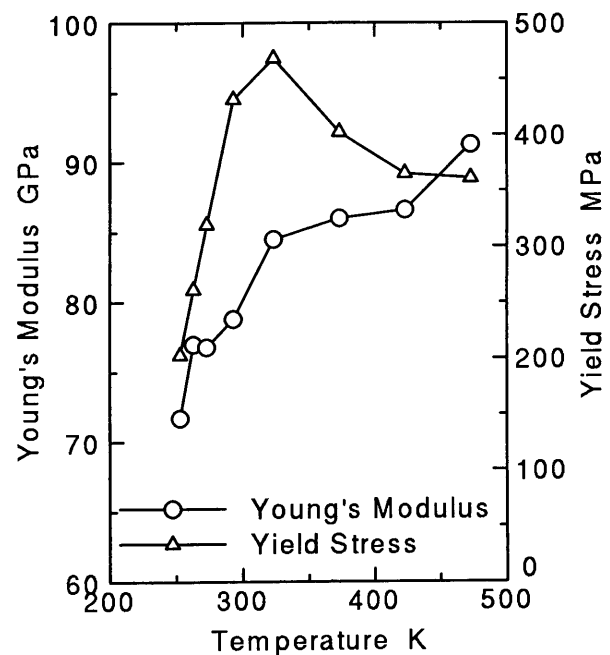


Fig. 5 Temperature dependency of Young's modulus and yield stress

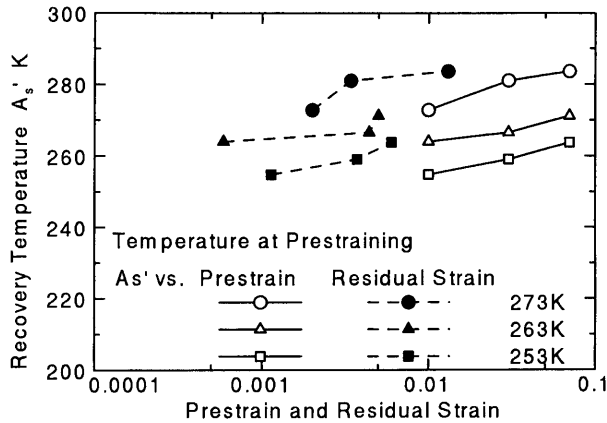


Fig. 6 Relation between A_s' and prestrain and residual strain

shows the relation between the temperature A_s' at which strain recovery starts and the prestrain and the residual strain. A_s' is defined as the temperature at which 0.001 of strain recovery is attained. Higher A_s' was observed for the higher preloading temperature. Also it is found that the larger prestrain results in the higher A_s' and the same trend is observed with the residual strain. The latter result coincides with the data reported by Duerig⁽⁴⁾.

For further discussion of the change of A_s' , it is correlated with the dissipated strain energy density W which is defined by the following equation.

$$W = \int_0^T \sigma \dot{\epsilon} dt \quad (1)$$

where T is the time when the unloading is completed. The result is shown in Fig. 7 from which the following empirical relation is obtained.

$$A_s' = A_s + (\theta - A_s)(1 - e^{-W/W_0}) + aW \quad (2)$$

where A_s and θ are the original phase transformation temperature and the pretraining temperature, respectively. $a = 6.9 \times 10^2 \text{ K}/(\text{J}/\text{mm}^3)$ and $W_0 = 3.3 \times 10^{-2} \text{ J}/\text{mm}^3$ are constants. Note that this equation gives $A_s' = A_s$ for $W = 0$.

5.2.2 Recovered strain and residual strain

Figure 8 shows a relation between the recovered and residual strains and the prestrain. Larger residual strain remains for larger prestrain at every temperature and a linear relation can be found between them. The similar result was reported by Duerig⁽⁴⁾. Also it can be estimated from Fig. 8 that there exists the minimum prestrain to get the strain recovery. They are determined from the figure as 0.00757 at 273 K (yield strain is 0.00423), 0.00482 at 263 K (yield strain is 0.00339) and 0.00419 at 253 K (yield strain is 0.00283). They are smaller than the yield strain and smaller for lower temperature. This may be important in application which uses the shape memory

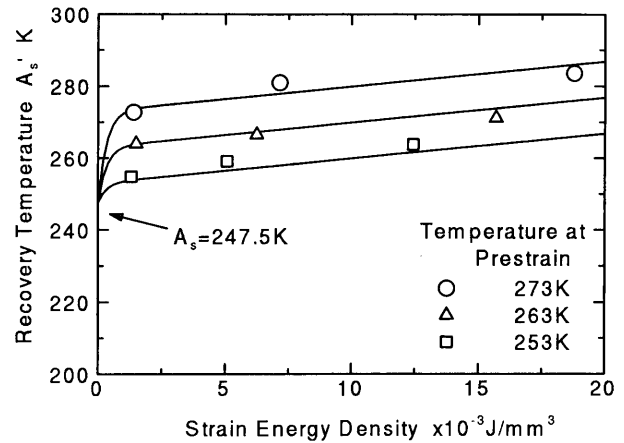


Fig. 7 Relation between A_s' and dissipated strain energy density

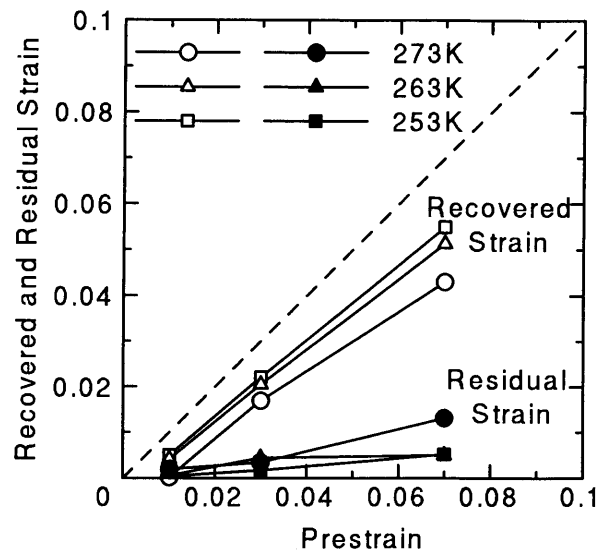


Fig. 8 Relation between recovered strain, residual strain and prestrain

effect and thus details should be investigated in future.

Figures 9(a), (b) and (c) show the recovered and residual strains as a function of pretraining temperature. For the prestrain of 0.01 and the temperature below 273 K, the residual strain is quite small and almost independent of the pretraining temperature. For the prestrain of 0.03 and 0.07 on the other hand, the higher temperature results in the larger residual strain. Finally, the higher temperature gives smaller strain recovery in all the cases.

The above discussion is summarized as follows. That is, by pretraining at lower temperature the larger strain recovery can be obtained. It suggests that in the application of the shape memory effect of the present SMA the pre-deformation should be given at low temperature. Otherwise the residual or plastic deformation can reduce the strain recovery effect.

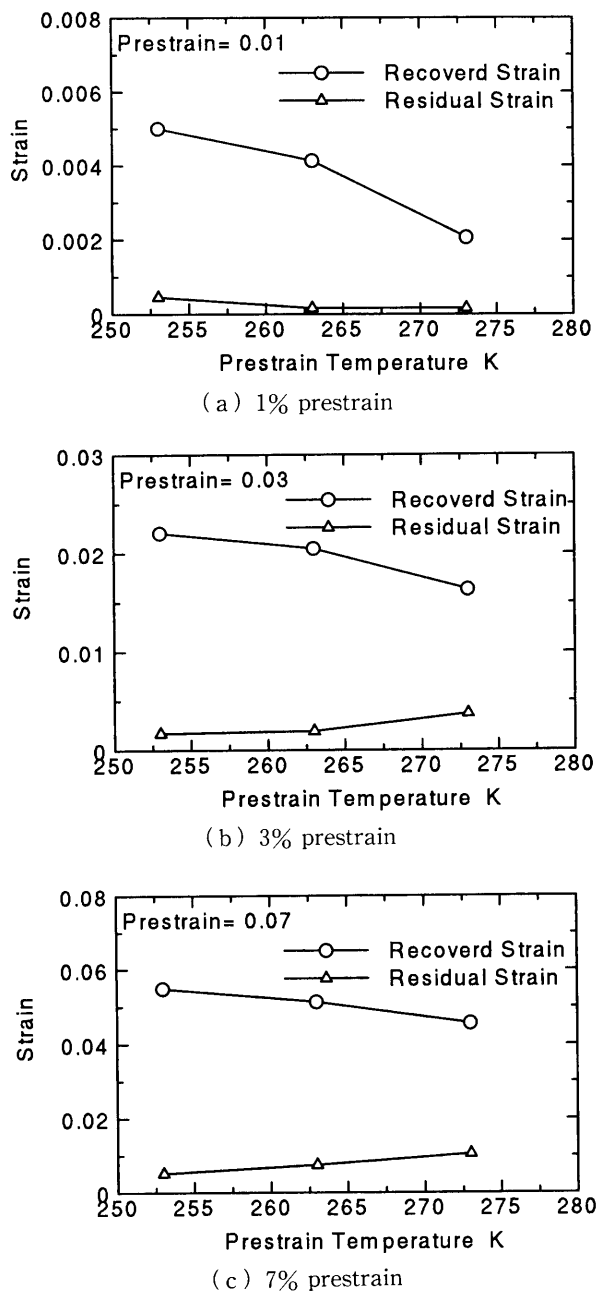


Fig. 9 Temperature dependence of recovered strain and residual strain

Also one should consider about the possible critical prestrain to observe the shape memory effect.

6. Conclusions

Fundamental deformation and recovery behaviors of Ni-Ti-Nb shape memory alloy were experimentally investigated. The experiments were conducted at the temperature range from 253 K to 473 K (200°C). The following conclusions are obtained.

(1) The Young's modulus is larger for higher temperature within the temperature range from 253 K (−20°C) to 473 K (200°C).

(2) The yield stress shows a linear positive relationship with temperature below 323 K (50°C) but shows the opposite trend above 323 K. The strain hardening is quite little below 323 K but is significant above 323 K.

(3) This indicates that the deformation mechanism is the stress induced martensite transformation below about 323 K but plastic deformation of the austenite phase above the temperature.

(4) The larger prestrain and the higher temperature results in higher A_s at which the strain recovery starts.

(5) There always observed the residual strain within the present temperature range.

(6) The lower prestraining temperature gives larger strain recovery.

(7) The existence of the minimum prestrain can be estimated to observe the shape memory effect. This critical value is smaller for lower temperature.

References

- (1) Tanaka, K., Tobushi, H. and Miyazaki, S., Mechanical Properties of Shape Memory Alloys, (in Japanese), (1993), Yokendo Ltd.
- (2) Fremont, M. and Miyazaki, S., Shape Memory Alloys, (1996), Springer.
- (3) Fujimaki, K., Hitoki, Y., Otsubo, T. and Uchiyama, J., Reliability Tests for Reactor Internals Rejuvenation Technology, Specialists Meeting on Behaviour of Core Internals, (1998), IAEA.
- (4) Duerig, T.W., Melton, K.N. and Proft, J.L., Wide Hysteresis Shape Memory Alloys, Duerig, T.W., Melton, K.N., Stoeckel, D., Wayman, C.M., eds., Engineering Aspects of Shape Memory Alloys, (1990), Butterworth-Heinemann.
- (5) Piao, M., Miyazaki, S. and Otsuka, K., Characteristics of Deformation and Transformation in Ti44Ni47Nb9 Shape Memory Alloy, Materials Transactions, JIM 33-4 (1992), pp. 346-353.
- (6) Zhang, C.S., Zhao, L.C., Duerig, T.W. and Wayman, C.M., Effects of Deformation on the Transformation Hysteresis and Shape Memory Effect in a NiTiNb Alloy, Scripta Metallurgica et Materialia, Vol. 24 (1990), pp. 1807-1812.
- (7) Zhao, L.C., Duerig, T.W., Justi, S., Melton, K.N., Proft, J.L., Yu, W. and Wayman, C.M., The Study of Niobium-Rich Precipitates in a Ni-Ti-Nb Shape Memory Alloy, Scripta Metallurgica et Materialia, Vol. 24 (1990), pp. 221-226.
- (8) Otsuka, K., Wayman, C.M., Nakai, K., Sakamoto, H. and Shimizu, K., Superelasticity Effects and Stress-Induced Martensitic Transformations in Cu-Al-Ni Alloys, Acta Metall., Vol. 24, (1976), pp. 207-226.
- (9) Otsuka, K., Sakamoto, H. and Shimizu, K., Successive Stress-Induced Martensitic Transformations and Associated Transformation Pseudoelasticity in Cu-Al-Ni Alloys, Acta Metall., Vol. 27

- (1979), pp. 585-601.
- (10) Abeyaratne, R. and Knowles, J.K., On the Driving Traction Acting on a Surface of Strain Discontinuity in a Continuum, *J. Mech. Phys. Solids*, Vol. 38, No. 3 (1990), pp. 345-360.
- (11) Abeyaratne, R. and Knowles, J.K., A Continuum Model of a Thermoelastic Solid Capable of Undergoing Phase Transitions, *J. Mech. Phys. Solids*, Vol. 41, No. 3 (1993), pp. 541-571.
- (12) Abeyaratne, R., Kim, S.-J. and Knowles, J.K., A One-Dimensional Continuum Model for Shape-Memory Alloys, *Int. J. Solids Structures*, Vol. 31, No. 16 (1994), pp. 2229-2249.
- (13) Abeyaratne, R. and Knowles, J.K., Unstable Elastic Materials and the Viscoelastic Response of Bars in Tension, *Trans. ASME, J. Applied Mechanics*, Vol. 55 (1988), pp. 491-492.
- (14) Abeyaratne, R., Kim, S.-J. and Knowles, J.K., Continuum Modeling of Shape-Memory Alloys, Brinson, L.C. and Moran, B. eds., *ASME AMD-189, PVP-292* (1994), pp. 59-69.
-